Reviewer Recommendation and Comments for Manuscript Number PONE-D-25-46962R3

Integrated single-cell RNA sequencing and Bulk-RNA technologies reveal the immunological characteristics of lactylation related-genes in glioblastoma

Revision Number 3
Willem Buys

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Recommendation: Minor Revision

Custom Review Question(s):	**Response**
Comments to the Author	All comments have been addressed
1. If the authors have adequately addressed your comments raised in a previous round of review and you feel that this manuscript is now acceptable for publication, you may indicate that here to bypass the "Comments to the Author" section, enter your conflict of interest statement in the "Confidential to Editor" section, and submit your "Accept" recommendation.	
2. Is the manuscript technically sound, and do the data support the conclusions? The manuscript must describe a technically sound piece of scientific research with data that supports the conclusions. Experiments must have been conducted rigorously, with appropriate controls, replication, and sample sizes. The conclusions must be drawn appropriately based on the data presented.	Partly
3. Has the statistical analysis been performed appropriately and rigorously?	Yes
4. Have the authors made all data underlying the findings in their manuscript fully available? The PLOS Data policy requires authors to make all data underlying the findings described in their manuscript fully available without restriction, with rare exception (please refer to the Data Availability Statement in the manuscript PDF file). The data should be provided as part of the manuscript or its supporting information, or deposited to a public repository. For example, in addition to summary statistics, the data points behind means, medians and variance measures should be available. If there are restrictions on publicly sharing data—e.g. participant privacy or use of data from a third party—those must be specified.	Yes
5. Is the manuscript presented in an intelligible fashion and written in standard English? PLOS ONE does not copyedit accepted manuscripts, so the language in submitted articles must be clear, correct, and unambiguous. Any typographical or grammatical errors should be corrected at revision, so please note any specific errors here.	Yes
6. Review Comments to the Author Please use the space provided to explain your answers to the questions above. You may also include additional comments for the author, including concerns about dual publication, research ethics, or publication ethics. (Please upload your review as an attachment if it exceeds 20,000 characters)	Reviewer Summary: In this revision of their primary research manuscript entitled "Integrated single-cell RNA sequencing and Bulk-RNA technologies reveal the immunological characteristics of lactylation related-genes in glioblastoma", Cheng et al. present a secondary analysis of lactate-related genes in published transcriptomics analyses of glioblastoma. Prior comments have mostly been addressed. I appreciate the authors opting for a more modest, conservative interpretation in their abstract, which better reflects the actual findings.

The rationale for this study is strong, the novelty and expected impact are moderately low, mainly owing to the depth of in silico analyses without adequate confirmation in vitro or in vivo.

Recommendation: Minor Revision.
Major Comments:
1. None.
Minor Comments:
2. Line 320: "suggesting" or "indicating" is likely more appropriate here than "confirming".
3. Line 383: "evaluated in silico" or "evaluated indicators of" is likely more appropriate here than simply "evaluated".
4. Line 446: Calling a model with an AUC of around 60% "reliable" is a stretch.
5. Please also discuss limitations of your work, such as the risk of overfitting and what steps were taken to address this risk.
6. You mention that CD93 were correlated with HLA molecules but was not affecting antigen presentation. This is interesting, as it is unexpected. Consider elaborating.
Language & Proofing:
7. The language quality in this manuscript is acceptable but shows room for improvement in clarity and expression.
8. Line 49: This sentence implies that machine learning designed a model, which might be slightly misleading wording. The authors designed a model informed by machine learning.
9. Lines 50-51: "The high-risk group had a high risk outcome." Pleonasm.
10. Line 51: "Immunological analysis" is suggestive of in vitro or in vivo experiments. However, I believe the authors are referencing in silico analyses of ex vivo data here. Consider slight rewording for accuracy.
11. Lines 80-81: "for tumor cells but tumor cells need a large amount of nutrients and oxygen for rapidly expanding tumor cells" This sentence may benefit from slight re-wording to avoid repeating "tumor cells" three times.
12. Line 141 still features "Role" instead of "Ro/e".
13. Line 293: There might be an omission here. What "resolution parameter" was used for this analysis?
14. Lines 430-431: I assume the authors mean "and confirmed" not "which confirmed".
15. Line 502: "confirmed ~~that~~ high". Consider taking out the word "that".
16. Lines 505 - 508: Circular reasoning.
Further Suggestions:
17. None.

7. PLOS authors have the option to publish the peer review history of their article (what does this mean?). If published, this will include your full peer review and any attached files.

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Yes: Willem Buys

Confidential to Editor

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2. Did you receive any assistance in preparing this review (e.g. from a post-doc or graduate student)? If yes, please include their name below.	
3. If accepted, do you think this submission should be highlighted on the PLOS ONE website? PLOS ONE does not evaluate manuscripts based on perceived significance or readership. We aim to provide tools for readers to filter and evaluate our publications. (optional)	No
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